

Module 4: Auto-Scaling Assignment

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Problem Statement: You work for XYZ Corporation that uses on premise solutions and some limited number of systems. With the increase in requests in their application, the load also increases. So, to handle the load the corporation has to buy more systems almost on a regular basis. Realizing the need to cut down the expenses on systems, they decided to move their infrastructure to AWS.

Tasks to do:

1. To Create a web server AMI with Apache server running in it.
2. To create a launch Template with this AMI.
3. To use this launch configuration to create an Auto Scaling group with 1 minimum and 3 maximum instances.

Task 1:

To Create a web server AMI with Apache server running in it.

A. Created EC2 instances

EC2 → Instances → Launch instance

Selected Configuration and launched 1 EC2 server.

ELB-Server-1

EC2 > Instances > Launch an instance

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-0fef201115eefe936 (64-bit (x86), uefi-preferred) / ami-0eb45f74aa8a20238 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

EC2 > Instances > i-03f191509e3878db6

Instance summary for i-03f191509e3878db6 (ELB-Server-1) Info

Updated less than a minute ago

Instance ID

i-03f191509e3878db6

IPv6 address

—

Hostname type

IP name: ip-172-31-11-238.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

3.239.18.96 [Public IP]

IAM role

—

IMDSv2

Required

Operator

—

Public IPv4 address

3.239.18.96 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-11-238.ec2.internal

Instance type

t3.micro

VPC ID

vpc-030682f2c586ad6b1

Subnet ID

subnet-02791cbf0e8268fb9

Instance ARN

arn:aws:ec2:us-east-1:430776404757:instance/i-03f191509e3878db6

Private IPv4 addresses

172.31.11.238

Public DNS

ec2-3-239-18-96.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses

—

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name

—

Managed

false

B. Installed Web-Servers in the EC2 servers

Connected to ELB-Server-1

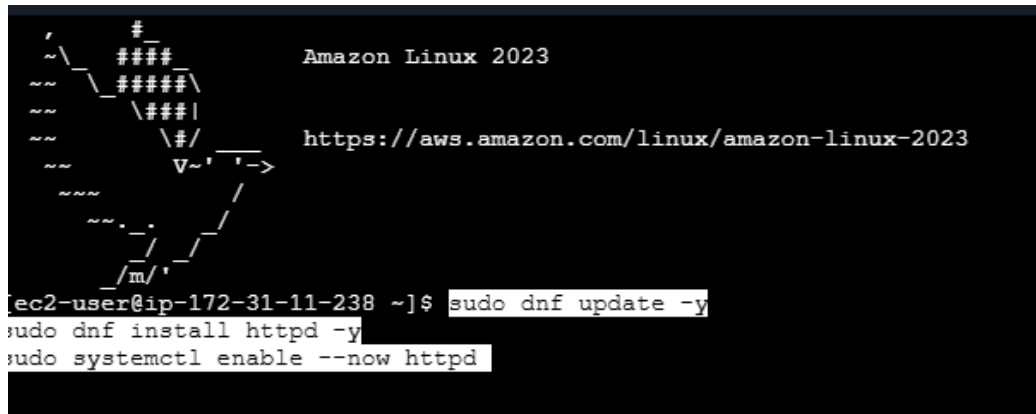
ELB-Server-1 → Connect

EC2 Instance Connect → Connect

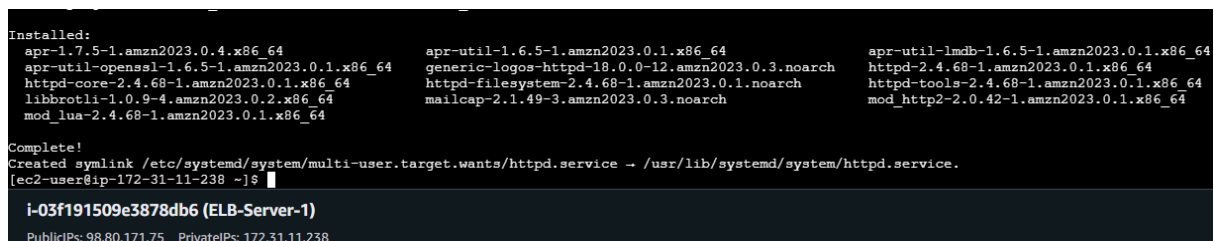
Ran the following bash to install Apache webserver and enable it:

On terminal ran:

```
sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
```

A terminal window on Amazon Linux 2023. The prompt is [ec2-user@ip-172-31-11-238 ~]\$. The user enters the commands: sudo dnf update -y, sudo dnf install httpd -y, and sudo systemctl enable --now httpd. The output shows the packages being installed and the service being enabled.

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-11-238 ~]$ sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
```

A terminal window showing the output of the installation. It lists the installed packages and the created symlink for the httpd service. The prompt is [ec2-user@ip-172-31-11-238 ~]\$.

```
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
httpd-core-2.4.68-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64
apr-util-1.6.5-1.amzn2023.0.1.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
mailcap-2.1.49-3.amzn2023.0.3.noarch
apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64
httpd-2.4.68-1.amzn2023.0.1.x86_64
httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
mod_http2-2.0.42-1.amzn2023.0.1.x86_64
Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-11-238 ~]$
```

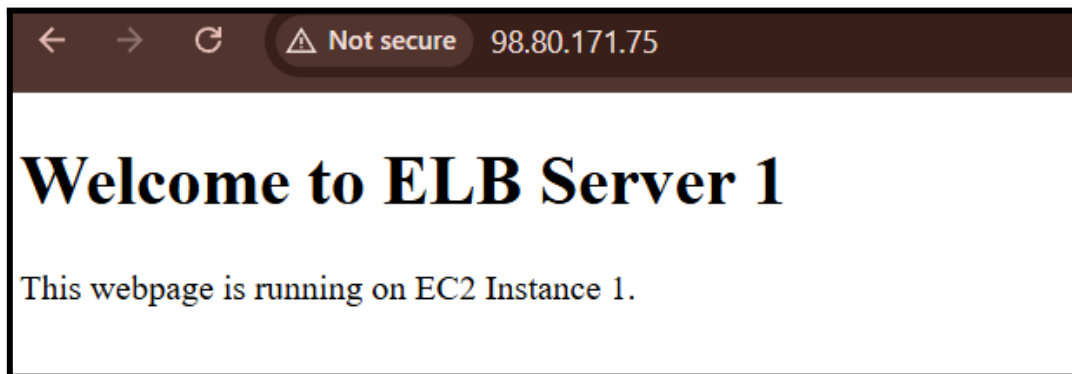
C. Created Server 1 webpage

Bash to open index.html : sudo nano /var/www/html/index.html

edited index.html to:

```
<!DOCTYPE html>
<html>
<head>
  <title>ELB Server 1</title>
</head>
<body>
  <h1>Welcome to ELB Server 1</h1>
  <p>This webpage is running on EC2 Instance 1.</p>
</body>
</html>
```

After installing the Apache webserver and editing, used the EC2 instance Public IP to check the webserver.

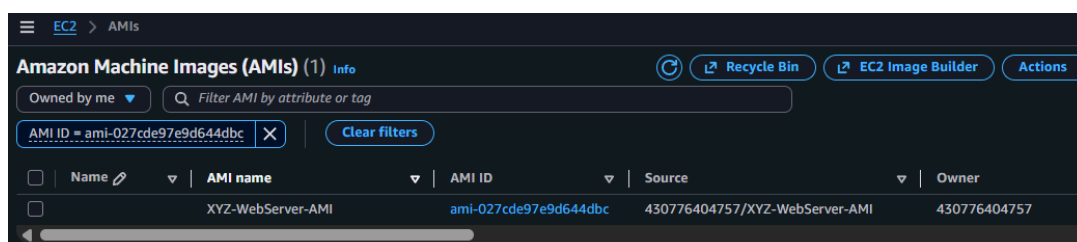
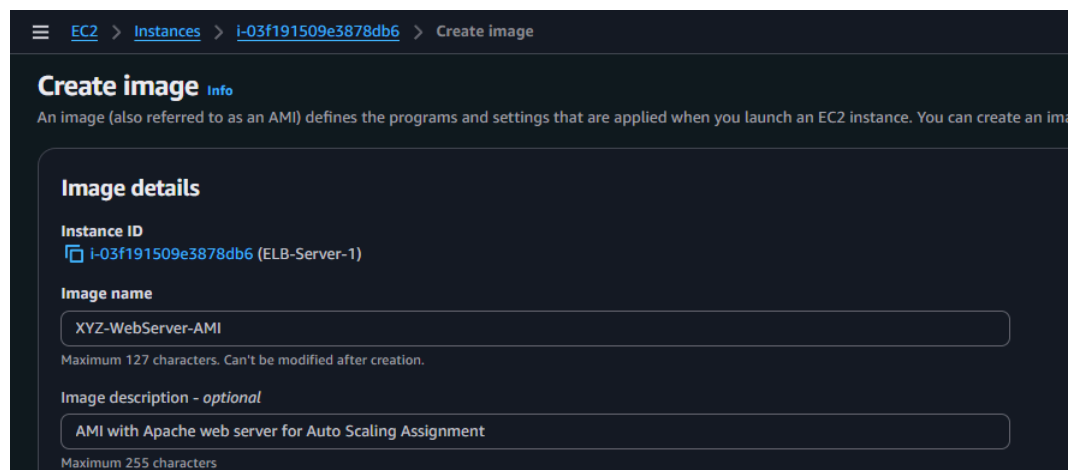


D. Created the AMI

1. Selected **ELB-Server-1**
2. Clicked **Actions**
3. Selected: **Image and templates** → **Create image**

Configured it as follows:

- Image name: **XYZ-WebServer-AMI**
- Image description: “**AMI with Apache web server for Auto Scaling Assignment**”



Task 2:

Created a launch Template with this AMI.

Steps and configurations :

EC2 → Launch Templates → Create launch template

Launch template name : **XYZ-WebServer-LaunchTemplate**

Template description : **Launch template for Auto Scaling Assignment**

AMI : My AMIs → Owned by me : **XYZ-WebServer-AMI** , **ami-027cde97e9d644dbc**

Instance type : **t3.micro**

Selected a keypair and left rest as default.

The screenshot shows the 'Create launch template' page in the AWS Management Console. The page is divided into two main sections: 'Launch template name and description' on the left and 'Summary' on the right. In the left section, the 'Launch template name' is 'XYZ-WebServer-LaunchTemplate' and the 'Template version description' is 'Launch template for Auto Scaling Assignment'. The 'Summary' section on the right lists the configuration: 'Software Image (AMI)' as 'AMI with Apache web server for...read more' (ami-027cde97e9d644dbc), 'Virtual server type (instance type)' as 't3.micro', 'Firewall (security group)' as '-', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. At the bottom right, there are 'Cancel' and 'Create launch template' buttons.

EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

XYZ-WebServer-LaunchTemplate

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@', or ' '.

Template version description

Launch template for Auto Scaling Assignment

Max 255 chars

Auto Scaling guidance [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

▶ Template tags

▶ Source template

Summary

Software Image (AMI)

AMI with Apache web server for...[read more](#)

ami-027cde97e9d644dbc

Virtual server type (instance type)

t3.micro

Firewall (security group)

-

Storage (volumes)

1 volume(s) - 8 GiB

Cancel Create launch template

The screenshot shows the 'Application and OS Images (Amazon Machine Image)' page in the AWS Management Console. It features a search bar at the top, followed by tabs for 'Recents', 'My AMIs', and 'Quick Start'. Under the 'My AMIs' tab, there are three filter buttons: 'Don't include in launch template', 'Owned by me' (which is selected), and 'Shared with me'. To the right, there is a 'Browse more AMIs' link. Below the filters, the 'Amazon Machine Image (AMI)' section displays details for 'XYZ-WebServer-AMI' (ami-027cde97e9d644dbc), including its creation date, virtualization type (hvm), ENA status, root device type (ebs), and boot mode (uefi-preferred). The 'Description' section shows 'AMI with Apache web server for Auto Scaling Assignment'. At the bottom, a table lists the 'Architecture' as 'x86_64' and the 'AMI ID' as 'ami-027cde97e9d644dbc'.

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

🔍 Search our full catalog including 1000s of application and OS images

Recents **My AMIs** Quick Start

☐ Don't include in launch template ☒ Owned by me ☐ Shared with me

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

XYZ-WebServer-AMI
ami-027cde97e9d644dbc
2026-09-28T04:52:51.000Z Virtualization: hvm ENA enabled: true
Root device type: ebs Boot mode: uefi-preferred

Description

AMI with Apache web server for Auto Scaling Assignment

Architecture	AMI ID
x86_64	ami-027cde97e9d644dbc

▼ Instance type

Info | Get advice

Advanced

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

mgja_keypair_1

Create new key pair

▼ Network settings

Info

Subnet

Info

Don't include in launch template

Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone

Info

Don't include in launch template

Enable additional zones

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Security groups

Info

Select security groups

Compare security group rules

► Advanced network configuration

XYZ-WebServer-LaunchTemplate (lt-025c633158d58857d)

Actions ▼

Delete template

Launch template details

Launch template ID

lt-025c633158d58857d

Launch template name

XYZ-WebServer-LaunchTemplate

Default version

1

Owner

arn:aws:iam::430776404757:root

Details

Versions

Template tags

Launch template version details

Actions ▼

Delete template version

Version

1 (Default)

Description

Launch template for Auto Scaling Assignment

Date created

2026-09-28T06:42:45.000Z

Created by

arn:aws:iam::430776404757:root

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID

ami-027cde97e9d644dbc

Instance type

t3.micro

Availability Zone

-

Availability Zone Id

-

Key pair name

mgja_keypair_1

Security groups

-

Security group IDs

-

Task 3:

To use this launch configuration to create an Auto Scaling group with 1 minimum and 3 maximum instances.

1. Created the Auto Scaling Group

AWS Console → EC2 → Auto Scaling Groups

Clicked Create Auto Scaling group

Group name: **XYZ-AutoScaling-Group**

Launch template: **XYZ-WebServer-LaunchTemplate**

Choose launch template Info

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name
Enter a name to identify the group.

XYZ-AutoScaling-Group

Must be unique to this account in the current Region and no more than 255 characters.

Launch template Info

ⓘ For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

XYZ-WebServer-LaunchTemplate

[Create a launch template](#)

Version
Default (2)

[Create a launch template version](#)

Description XYZ-WebServer-LaunchTemplate	Launch template XYZ-WebServer-LaunchTemplate lt-025c633158d58857d	Instance type t3.micro
AMI ID ami-027cde97e9d644dbc	Security groups -	Request Spot Instances No
Key pair name mgia_keypair_1	Security group IDs sg-04191fc6a271cf3a4	

Additional details

Storage (volumes) -	Date created Mon Sep 28 2026 12:23:53 GMT+0530 (India Standard Time)
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Scaling

Info

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

Max desired capacity

1

3

Equal or less than desired capacity

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy

Info

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

XYZ-AutoScaling-Group

XYZ-AutoScaling-Group Capacity overview

Edit

arn:aws:autoscaling:us-east-1:430776404757:autoScalingGroup:26b28c4d-98a5-4270-a7c2-a38353f57360:autoScalingGroupName/XYZ-AutoScaling-Group

Desired capacity	Scaling limits	Desired capacity type	Status
1	1 - 3	Units (number of instances)	Updating capacity

Date created
Mon Sep 28 2026 12:43:15 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Edit

Launch template lt-025c633158d58857d XYZ-WebServer-LaunchTemplate	AMI ID ami-027cde97e9d644dbc	Instance type t3.micro	Owner arn:aws:iam::430776404757:root
Version Default	Security groups -	Security group IDs sg-04191fc6a271cf3a4	Create time Mon Sep 28 2026 12:23:53 GMT+0530 (India Standard Time)
Description XYZ-WebServer-LaunchTemplate	Storage (volumes) -	Key pair name mgia_keypair_1	Request Spot Instances No

View details in the launch template console

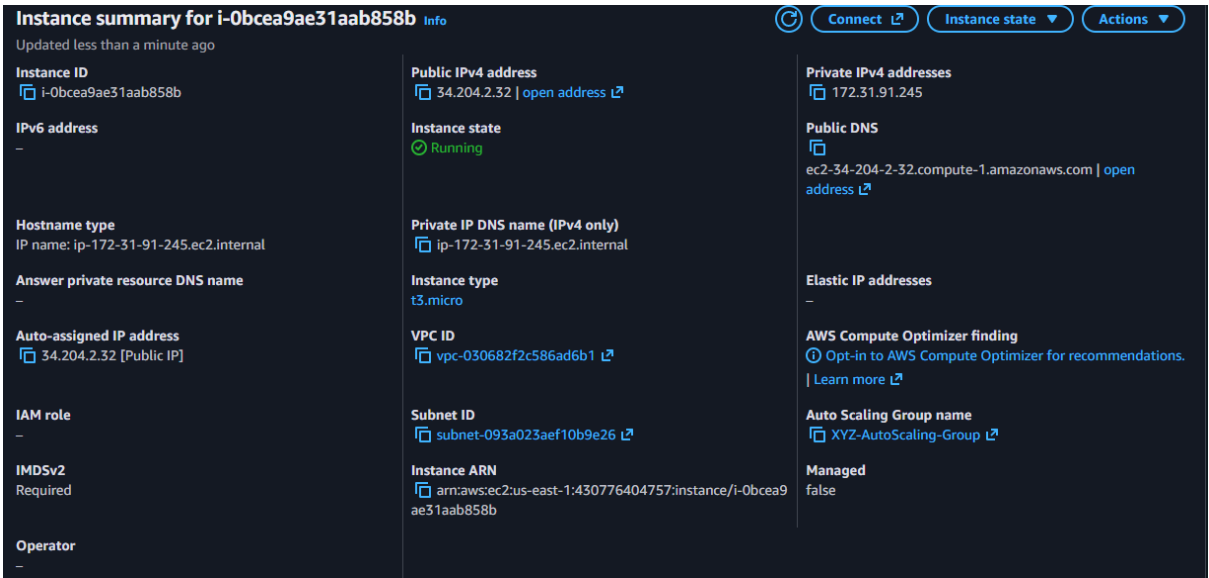
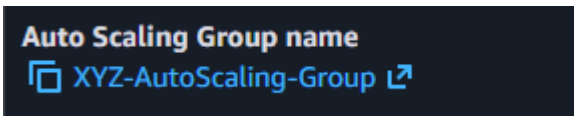
Network

Edit

Availability Zones use1-az2 (us-east-1b) use1-az1 (us-east-1a)	Subnet ID subnet-093a023aef10b9e26 subnet-02791cbf0e8268fb9	Availability Zone distribution Balanced best effort
--	---	--

Successfully created autoscaling-group capacity 1-3 instances.

Verified the newly created instance



Used Public IP to check - webserver successfully launched.



